

Large ρ does not make progressive hedging diverge on the farmer problem; it makes the convergence metric lie

David L. Woodruff

August 30, 2026

Abstract

Progressive hedging (PH) is usually reported as converged when the primal residual $\|x_s - \bar{x}\|_1$ falls below a threshold. We ask whether a very large penalty parameter ρ makes that metric increase — whether PH diverges — on the three-crop, three-scenario farmer problem. It does not. For $\rho \geq 10^4$ the metric reaches solver tolerance at the second iteration and never rises again, while $\bar{x}$ sits 58 acres from the extensive-form solution and creeps toward it at a rate of exactly $172.6/\rho$ acres per iteration. We explain the freeze: when $\bar{x}$ is feasible for the first-stage constraints, a dominant proximal term makes every subproblem return $\bar{x}$ itself, and the PH iteration degenerates into a projected gradient step of length $1/\rho$ on the extensive-form objective. The metric does increase, by as much as a factor of 5.4×10^6 in a single iteration, but at the intermediate values $\rho \in \{100, 1000\}$ rather than at the large ones. FWPH behaves identically, to six figures, because a dominant proximal term makes its Frank-Wolfe columns irrelevant. On this problem the primal residual measures agreement among subproblems, not proximity to a solution, and the two are easy to tell apart only if one is willing to solve the extensive form.

1 Question and setup

Let S index scenarios with probabilities p_s , let $x_s \in \mathbb{R}^n$ be the first-stage variables of scenario s , and let $\bar{x} = \sum_s p_s x_s$. mpi-sppy's `phbase.convergence.diff` reports

$$\text{conv} = \frac{1}{|S|} \sum_{s \in S} \frac{1}{n} \|x_s - \bar{x}\|_1, \quad (1)$$

and terminates PH when this falls below `convthresh`. The question is whether (1) increases when ρ is made very large.

We use the farmer problem [1] in the form distributed with mpi-sppy, `examples/farmer/farmer.py`, with `crops_multiplier` = 1 (three crops: wheat, corn, sugar beets) and three scenarios (below-average, average and above-average yields), continuous first-stage acreage variables $x \in \mathbb{R}_+^3$ subject to the single constraint $\sum_j x_j \leq 500$. The extensive form solves to

$$x^* = (\text{corn } 80, \text{ sugar beets } 250, \text{ wheat } 170), \quad z^* = -108390.$$

Runs use the PH hub alone — no spokes, no bound-producing cylinders, one MPI rank — so that nothing outside the PH iteration can influence the trajectory. Each run is 100 iterations at a fixed ρ with `convthresh` set to `-1` so that no run stops early, solved with Gurobi through `gurobi_direct`. The metric (1) is recorded by a PH extension whose `miditer()` callback fires immediately after `phbase` assigns `self.conv`, so the recorded sequence is exactly the one `phbase` would have compared against the threshold.

ρ	conv ₁₀₀	rises	max jump	$\ \bar{x} - x^*\ _\infty$	drift
0.1	1.0×10^{-1}	38	4.8	0.381	5.2×10^{-1}
1	2.6×10^{-4}	38	4.0×10^1	0.00097	7.0×10^{-4}
10	3.7×10^{-3}	35	3.9×10^5	0.0139	7.9×10^{-3}
10^2	8.1×10^{-9}	46	5.4×10^6	32.0	1.3×10^{-1}
10^3	9.5×10^{-8}	39	3.5×10^5	47.7	4.3×10^{-2}
10^4	8.4×10^{-8}	2	1.2	56.6	1.7×10^{-2}
10^5	2.1×10^{-8}	0	1	58.2	1.7×10^{-3}
10^6	1.5×10^{-8}	0	1	58.3	1.7×10^{-4}
10^8	5.6×10^{-12}	0	1	58.3	1.7×10^{-6}

Table 1: PH on the three-crop farmer, 100 iterations, threshold disabled. “conv₁₀₀” is (1) at the last iteration; “rises” counts iterations after the first at which it increased; “max jump” is the largest single-iteration increase as a ratio; the fifth column is the true distance from the final $\bar{x}$ to the extensive-form solution; “drift” is how far $\bar{x}$ still moved on the last iteration.

2 Results

Table 1 sweeps ρ over nine values spanning nine orders of magnitude.

Three things are visible.

At large ρ the metric does not rise at all. For $\rho \geq 10^5$ the “rises” column is zero: (1) is monotonically non-increasing over the entire run. It reaches 10^{-8} — Gurobi’s feasibility tolerance, not a meaningful number — at iteration 2 and stays there. Any threshold a user would plausibly set is met almost immediately.

The reported convergence is spurious. At $\rho = 10^5$ the metric reads 2×10^{-8} while $\bar{x}$ is 58 acres per crop away from x^* ; the run has planted 308 acres of sugar beets where the answer is 250. The fifth column is monotone in ρ over the whole sweep and saturates at 58.3, which is the distance from x^* to the mean of the three scenarios’ individual wait-and-see solutions. In other words, at large ρ PH never leaves its starting point.

The metric rises in the middle of the range. At $\rho = 100$ and $\rho = 1000$ it repeatedly falls to 10^{-10} and jumps back to order one: the largest single-iteration increase is a factor of 5.4×10^6 at $\rho = 100$. These are the spikes in the left panel of Figure 1. Each spike is $\bar{x}$ drifting far enough that one scenario’s linear program flips to a different vertex; the three scenarios disagree for a few iterations, and the proximal term then pins them back together. The run is nonetheless making progress — the right panel shows the true error falling throughout — so the non-monotonicity of (1) is not divergence either.

3 Why it freezes

Write the PH subproblem for scenario s at iteration k as

$$x_s^{k+1} \in \arg \min_{x \in X} c^\top x + Q_s(x) + (w_s^k)^\top x + \frac{\rho}{2} \|x - \bar{x}^k\|_2^2, \quad (2)$$

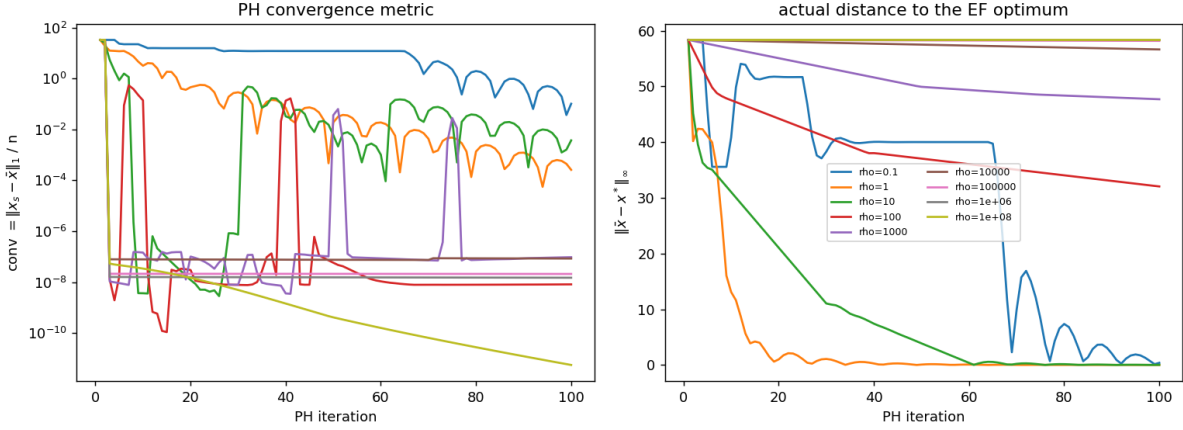


Figure 1: Left: the convergence metric (1) against iteration, log scale. Right: $\|\bar{x} - x^*\|_\infty$ against iteration, linear scale. The curves that flatten near the top of the right panel are the large- ρ runs whose metric, on the left, has already reached 10^{-8} .

where $X = \{x \geq 0 : \sum_j x_j \leq 500\}$ is the first-stage feasible set and Q_s is the scenario recourse function. The essential structural fact about farmer is that $\bar{x}^k \in X$ for every k : X is convex and $\bar{x}^k$ is a convex combination of points of X . As $\rho \rightarrow \infty$ the quadratic term dominates, and the minimizer of (2) approaches the projection of $\bar{x}^k$ onto X — which is $\bar{x}^k$ itself. Every scenario returns the same point, the residual (1) is zero up to solver tolerance, and PH declares victory.

The motion that remains is a first-order correction. Expanding the optimality condition of (2) about $\bar{x}^k$,

$$x_s^{k+1} \approx \Pi_X \left(\bar{x}^k - \frac{1}{\rho} \left(c + w_s^k + g_s^k \right) \right), \quad g_s^k \in \partial Q_s(\bar{x}^k), \quad (3)$$

and averaging over scenarios, using the PH invariant $\sum_s p_s w_s^k = 0$,

$$\bar{x}^{k+1} \approx \Pi_X \left(\bar{x}^k - \frac{1}{\rho} \left(c + \sum_s p_s g_s^k \right) \right). \quad (4)$$

The bracket is a subgradient of the extensive-form objective at $\bar{x}^k$. So at large ρ , PH is not diverging and is not stalling in any exotic sense: it has degenerated into projected subgradient descent on the extensive form with a fixed step length $1/\rho$. The dual variables w_s have stopped doing anything, because they only grow through $\rho(x_s - \bar{x})$ and $x_s - \bar{x}$ has been driven to zero.

Equation (4) predicts that the per-iteration motion of $\bar{x}$ should scale exactly as $1/\rho$, and it does. The measured drift multiplied by ρ is

$$172.556, \quad 172.556, \quad 172.555, \quad 172.6$$

at $\rho = 10^4, 10^5, 10^6, 10^8$ — constant to five figures over four decades. Since the gap to close is 58.3 acres, reaching x^* requires about $58.3 \rho / 172.6 \approx 0.34 \rho$ iterations: roughly 34,000 at $\rho = 10^5$ and 3.4×10^7 at $\rho = 10^8$. The algorithm is correct and hopeless at the same time.

4 FWPH does the same thing

Frank-Wolfe PH [2] replaces the exact subproblem solve with an inner Frank-Wolfe loop over a growing set of columns, and it carries its own convergence check — Algorithm 3 of Boland

et al., $\sum_s p_s \|x_s^{QP} - \bar{x}\|_2^2$, a probability-weighted squared two-norm not scaled by the variable count. It also reports an outer bound, which PH does not. Running the same instance and the same sweep gives Table 2.

ρ	conv ₁₀₀	rises	bound ₁	bound ₁₀₀	$\ \bar{x} - x^*\ _\infty$
0.1	5.5×10^{-2}	28	-115406	-108391	0.428
1	1.2×10^{-7}	36	-115406	-108390	0.00064
10	5.8×10^{-5}	31	-115406	-108392	0.0139
10^2	1.0×10^{-17}	5	-115406	-112258	32.0
10^3	1.6×10^{-23}	4	-115406	-112978	47.7
10^4	3.1×10^{-24}	0	-115406	-115406	56.6
10^5	3.2×10^{-24}	0	-115406	-115406	58.2
10^6	3.7×10^{-24}	0	-115406	-115406	58.3
10^8	3.4×10^{-25}	0	-115406	-115406	58.3

Table 2: FWPH on the same instance, 100 major iterations, ten Frank-Wolfe steps each, threshold disabled. “rises” ignores changes below conv = 10^{-12} , beneath which the squared metric is machine zero and its motion is solver noise. bound₁ and bound₁₀₀ are the FWPH outer bound at the first and last major iteration.

The conclusion is the same and the numbers are, to six figures, literally the same. For $\rho \geq 10^4$ the metric never rises, the outer bound does not move off its iteration-1 value at all, and the final $\bar{x}$ is the same 58.3 acres from x^* . The measured drift times ρ is 172.5556, 172.5556, 172.556, 172.6 at $\rho = 10^4 \dots 10^8$ — agreeing with the PH figures of Section 3 to every digit either run resolves.

That agreement is what (4) predicts. The Frank-Wolfe inner loop searches over convex combinations of columns for a point that minimizes the subproblem objective; when the proximal term dominates, that minimizer is $\bar{x}$ regardless of which columns are available, so the columns stop mattering and the major iteration reduces to the same projected gradient step of length $1/\rho$. Large ρ does not merely fail to be helped by Frank-Wolfe, it makes the Frank-Wolfe machinery inert.

Two differences from PH are worth recording. The metric being squared roughly doubles the exponent of everything: the spikes at $\rho = 10$ to 10^3 reach ratios of 10^{12} to 10^{13} rather than 10^6 , and the floor sits near 10^{-24} rather than 10^{-8} . And the outer bound is a second, independent witness that PH does not have: at $\rho \geq 10^4$ it is frozen at the iteration-1 value to every digit printed, so a user watching the bound rather than the residual would also see nothing wrong — only nothing happening.

5 Discussion

The practical reading is that on this problem (1) is a measure of agreement among subproblems and not a measure of optimality, and that large ρ decouples the two completely. This is worse than divergence would be. A diverging metric is visible; a metric that reads 10^{-8} while the iterate is 58 acres from the answer is not, and it will silently satisfy any termination test built on it alone.

Two caveats bound the claim. First, the freeze depends on $\bar{x}^k \in X$, which holds here because the first stage has a single knapsack-like constraint and no integrality. With integer first-stage variables $\bar{x}$ is generally infeasible for the subproblems and the mechanism of (4) does not apply unchanged; whether the metric then rises is a separate experiment. Second, the sweep is one instance at one scenario count, and the intermediate ρ behavior in particular — where the vertex flips live — should be expected to depend on the geometry of the recourse function.

The obvious defense is to not rely on the primal residual alone. A dual residual, $\rho \|\bar{x}^{k+1} - \bar{x}^k\|$, is exactly the quantity (4) says is constant at 172.6 here, and would have flagged all four large- ρ runs as unconverged. That this is available essentially for free suggests it belongs in the default termination test rather than beside it.

6 Reproducing

Everything is in `examples/farmer/divergence` on the `divergence_study` branch:

<code>farmer_ph_divergence.py</code>	PH driver; sweeps ρ , writes the table, csv and figure
<code>farmer_divergence.csv</code>	per-iteration ρ , conv, $\ w\ _\infty$, error, $\bar{x}$
<code>farmer_divergence.png</code>	Figure 1
<code>farmer_fwph_divergence.py</code>	FWPH driver; Table 2
<code>farmer_fwph_divergence.csv</code>	per-iteration conv, outer bound, error, $\bar{x}$
<code>notes.md</code>	the same findings in brief

`python farmer_ph_divergence.py --itermax 100` reproduces Table 1.

References

- [1] J. R. Birge and F. Louveaux, *Introduction to Stochastic Programming*, 2nd ed., Springer, 2011.
- [2] N. Boland, J. Christiansen, B. Dandurand, A. Eberhard, J. Linderoth, J. Luedtke and F. Oliveira, Combining progressive hedging with a Frank–Wolfe method to compute Lagrangian dual bounds in stochastic mixed-integer programming, *SIAM Journal on Optimization* 28(2):1312–1336, 2018.
- [3] R. T. Rockafellar and R. J.-B. Wets, Scenarios and policy aggregation in optimization under uncertainty, *Mathematics of Operations Research* 16(1):119–147, 1991.